(i) The formula for the area of an oblique triangle says that the area is equal to one half of the product of two sides and the sine of their included angle.

(ii) The area formula is used to compute the area of the segment of a circle. Segment of a circle is the region between a chord and the arc it cuts off.

Let r be the radius of a circle and θ be the angle subtended by the chord AB at the centre.

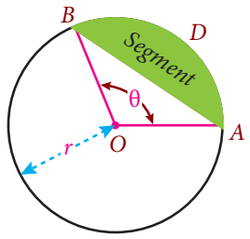


Figure 3.22

$$\text{Area of the segment } ABD = \text{Area of the sector} - \text{Area of the } \triangle OAB$$

$$= \frac{1}{2}r^2\theta - \frac{1}{2}r^2 \sin\theta$$

$$= \frac{1}{2}r^2 (\theta - \sin\theta)$$

(iii) The area formula of a triangle is viewed as generalisation of area formula of a right triangle.

(iv) In the above formula, it is clear that the measure of third side is not required in finding the area of the triangles. Also there is no need of finding the altitude of the triangle in order to find its area.

Example 3.56

The Government plans to have a circular zoological park of diameter $8\ km$. A separate area in the form of a segment formed by a chord of length $4\ km$ is to be allotted exclusively for a veterinary hospital in the park. Find the area of the segment to be allotted for the veterinary hospital.

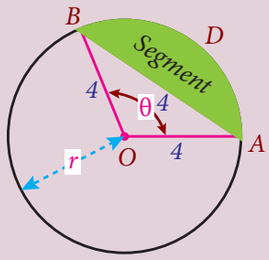


Figure 3.23

Solution:

Let AB be the chord and O be the centre of the circular park.
Let $\angle AOB = \theta$.

$$\text{Area of the segment} = \text{Area of the sector} - \text{Area of} \triangle OAB.$$

$$= \frac{1}{2}r^2\theta - \frac{1}{2}r^2 \sin\theta$$

$$= \left(\frac{1}{2} \times 4^2\right)[\theta - \sin\theta] = 8\,[\theta - \sin\theta] \qquad \text{...(i)}$$

$$\text{But } \cos\theta = \frac{4^2 + 4^2 - 4^2}{2(4)(4)} = \frac{1}{2}$$

$$\text{Thus, } \theta = \frac{\pi}{3}$$

From (i), area of the the segment to be allotted for the veterinary hospital

$$= 8\left[\frac{\pi}{3} - \frac{\sqrt{3}}{2}\right] = \frac{4}{3}\left[2\pi - 3\sqrt{3}\right]\ m^2$$